Version 1.0 (Jan 2023) | Written by: Ze Lei Tan, Vetted by: Dr Furene Wang, Prof Ong HT (Paeds Neuro), Dr Kristy Fu (PICU), Dr Ian Tan (Children's Emergency)

Paediatric Stroke: Assessment and Activation Pathway in ED

Focal neurological deficits: hemiparesis/hemifacial weakness, aphasia, visual disturbances, ataxia, **± any:**
a) Non-localizing symptoms: headache, altered mental status
b) Raised intracranial pressure: Cushing's reflex, Lateral gaze palsy, anisocoria, papilledema
c) Known conditions: Cardiac disease, cerebral arteriopathy, autoimmune disease

Urgent Assessment and Stabilization

1. Neurologic assessment with **PedNIHSS** (under "Children's Emergency" or "Paediatric Critical Care".
2. Obtain venous access (minimum 22G):
   - POCT glucose, micro-electrolytes
   - FBC, DIC screen, GXM, Urea / Cr / Electrolytes
   - Other invx depend on clinical concerns (if there is suspected ICH, refer below right)
3. Medical stabilization
   - Secure airway if needed
   - Neuroprotective measures
4. Consider nasogastric tube ± Urinary catheter

Paediatric Stroke Team*
1. Paediatric Neurologist on call
2. Adult Acute Stroke Team on call
3. Paediatric ICU con on call

Suggested communication:
"This is a Paeds AIS stroke activation. PedNIHSS score is ___. CTB and CTA ongoing / CTB and CTA neg / concerning for ischemic stroke."

For EVT:
1. Interventional neuroradiologist (Dr Anil Gopinathan 97296614/Dr Yang Cunli 91595700)

1. Record time of onset of symptoms
2. Record PedNIHSS score
3. Activate Paediatric Stroke Team*
4. Stat CT brain + CTA
Discuss with Stroke team re: MRI/MRA if time of onset is unclear

Intracranial Haemorrhage

In CE
1. Activate PICU, Paeds Neuro, NES
2. IV Access: FBC, DIC screen, GXM, Consider ROTEM [ECMO

In PICU, consider central access
1. Before transfusions:
- 1 x EDTA tube for DNA extraction (ice)
- 1 x Li Hep tube: Plasma aa + ac (ice)
- Spot urine organic acids (ice)

Discuss with various subspecs re: i) Viral titres ii) Blood spot carnitine iii) Blood Cu and Ceruloplasmin iv) factor assays

2. When blood draw is possible: Ca, Phos, ALP, PTH, Vit D if skull #

No Intracranial Haemorrhage
Suspect ischemic stroke

Clinical criteria:
1. ≥ 2yo
2. Neurologic deficit + pedNIHSS score **≥ 6 , ≤ 24**

Neuroimaging criteria:
1. No intracranial haemorrhage
2. Signs of ischemia consistent with large vessel occlusion (anterior and posterior circulation territories)

Symptom onset ≤ 4.5h

If no contraindications, Consider IV TPA in ED ± EVT #

Symptom onset >4h, ≤ 6h

If no contraindications, consider EVT #

Neuroprotective measures
Seizure prophylaxis (Keppra 10mg/kg BD)
PICU admission

Symptom onset >6h

Consult stroke team

IF NO BLEED + not for TPA/EVT, consider:
i) ASA 5mg/kg after 24 hours
ii) Neuroprotective measures
iii) Seizure prophylaxis Keppra 10mg/kg BD
iv) PICU admission

Within the shaded portion of this protocol, all decisions to be discussed with Paediatric Stroke Team*

To review contraindications, refer to Pg 2 below

Hyperacute Recanalization Therapy

Contraindications

a. Thrombolysis (IV TPA)

i. Brain/other vascular malformations
ii. Brain tumour
iii. Congenital or acquired bleeding diatheses
iv. Thrombocytopenia
vi. Any thrombolysis / anticoagulation administered in the last 24-48 hrs
vii. Gastrointestinal / genitourinary tract bleeding in the preceding 21 days
viii. Haemorrhage seen on CT Brain/MRI Brain, or haemorrhage elsewhere
ix. Extensive infarct size on initial CT scan

b. Endovascular Thrombectomy – EVT (relative)

i. Small child with accompanying technical limitations
ii. Suspicion of underlying vasculitis (eg. FCA) or arteriopathy (eg. Moyamoya)
iii. Serious sensitivity to radiological contrast/nickel, titanium metals or their alloys
iv. Significant acute kidney injury / chronic kidney disease (risk of contrast-induced nephropathy)

1. IV thrombolysis (IV rt-PA) checklist (Acute Management)

1. **If there are no contraindications, IV TPA dose: 0.9mg/kg (max 90mg) – ordered and given by Stroke Team**
   a. Administer 10% of bolus in 5 min, remaining 90% in the next 1 hour
   b. IV rt-PA can be commenced in the CE after discussion with various subspecialty teams (ref. Pg 1).
2. Monitoring
   a. Close blood pressure monitoring:
      - Every 15 minutes for 2 hours after commencement of IV TPA
      - Every 30 minutes for 6 hours
      - Every 60 minutes until 24 hours after IV TPA dose is commenced
      - Aim for normotension (50-90th percentile for age)
   b. Neurological monitoring
      - Ped NIHSS
        > Before administration of rt-PA
        > 1h after completion of rt-PA infusion
      - Hourly GCS chart
3. Watch for potential complications
   a. Intracranial haemorrhage
      - If the patient has severe headache / Ped NIHSS score increases by 4 or more / if GCS drops by 2 or more:
        > Discontinue rt-PA infusion if it is still ongoing
        > Send FBC, DIC screen, GXM, Consider ROTEM
        > Arrange urgent CT Brain
        > Obtain neurosurgical consult if necessary
        > Correct underlying platelet or coagulation abnormalities
   b. Non-CNS bleeding complications
      - Treat minor oozing with pressure dressing
      - Consider correcting underlying platelet or coagulation abnormalities
      - Regardless of the presence /absence of bleeding,
        > No antiplatelet /anticoagulation for 24 hours from start of TPA
        > Avoid invasive procedures

c. Anaphylaxis – to treat as such. Refer to "Management of Anaphylaxis in Paediatric Patients" under our Paediatric Allergy, Immunology and Rheumatology guidelines.

2. Endovascular Thrombectomy (EVT) checklist

1. Paediatric neurologist and interventional neuroradiologist confirm patient eligibility for EVT, to be done under GA in IR / Angio Suite
2. Activate **Anaesthetist on call**
3. Activate "**'Admit to Inpatient"** order on EPIC
4. Liaise with IR / Angio suite when ready to receive case
5. Liaise with PSA to transfer patient to "NWEDS" virtual ward system when patient is being transferred to IR/Angio Suite
6. Monitoring:
   a. As per Anaesthesia
   b. Neurological monitoring
      - Ped NIHSS score before and after EVT
      - Transfer to PICU – refer to point 3 for post recanalization therapy monitoring
7. Disposition: PICU

3. Management principles post recanalization therapy in the PICU

1. **Neuroprotective measures**
   a. Nurse head up 30deg
   b. Normothermia
   c. Normotension (50-90th percentile for age)
   d. Respiratory support, Spo2 95-99%
   e. Na aims 140-145mmol/L, normocarbia 35-40mmHg
   f. 3% NaCl 3-5ml/kg
2. **Choice of fluid: 0.9% Na tonicity for the first 24 hours**
3. **BP monitoring**
4. **Close neuromonitoring:**
   a. Ped NIHSS
      i. Before and after recanalization therapy
      ii. 1 hr after recanalization therapy
      iii. Once per shift for first 24 hours
      iv. Once daily for duration of PICU / PHD stay thereafter
   b. Hourly GCS chart for duration of PICU / PHD stay
   c. If Ped NIHSS increases by 4 or more / GCS drops by 2 or more / severe headache / neurologic deficit, considerations:

      i. Risk of worsening cerebral edema / Malignant MCA infarction in the first 24-72h
      ii. Send FBC, DIC screen, GXM, Consider ROTEM
      iii. Arrange urgent CT Brain
      iv. Obtain neurosurgical consult if necessary
      v. Correct underlying platelet or coagulation abnormalities

5. **Clinical and electrical seizure monitoring**
6. **Consider seizure prophylaxis (Keppra 10mg/kg q12hr)**
7. **Monitor cannulation site: bleeding and perfusion**
8. **Repeat imaging 24h after recanalization therapy, or sooner if there is neurologic deterioration (Refer point 3); (Discuss with Neuro, IR)**